

Polynomial Regression Assignment

Roll number: BT2024175

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GitHub: github.com/Jos-Samuel/BT2024175-polynomial-regression

The final models were degree-5 LASSO for var1 ($\alpha = 0.01$) and degree-11 Elastic Net for var2 ($\alpha = 0.001$, L1 ratio = 0.3). The following sections describe the preprocessing, parameter search and validation results used to make these choices.

1. Objective and data

The assignment provides two datasets. In var1, the target is Net Power Score and the six inputs describe turbine operating adjustments. In var2, the target is Thermal Anomaly Score and the three inputs are spatial coordinates. Each dataset was fitted separately, using only its own training observations.

Property	var1: turbine	var2: reservoir
Training / test rows	1,000 / 1,000	1,000 / 1,000
Input variables	x1 to x6	x1 to x3
Candidate degrees	1 to 10	1 to 20
Data checks	Complete; 0 repeats*	Complete; 12 repeats*

Complete means no missing values. *Repeats count rows beyond the first occurrence of an identical training input vector. Training files contain y; test files contain inputs only.

2. Polynomial features and preprocessing

For degree d , all monomials with total degree at most d were included, including interactions and lower-order terms. For example, a degree-2 expansion of two inputs includes x_1 , x_2 , x_1^2 , x_1x_2 and x_2^2 . A separate constant feature was omitted because the intercept is handled by the regression step.

Pipeline: standardize raw inputs → generate polynomial terms → standardize polynomial columns → fit the regularized model.

Each scaler was fitted on the current training fold only. Validation and test rows were transformed using those fitted means and standard deviations. This prevents information from held-out rows influencing training. The supplied inputs already lie within $[-1, 1]$; the first scaling step was retained consistently, while scaling the expanded columns makes regularization less dependent on differences in monomial variance.

With p inputs, the expansion contains $C(p + d, d) - 1$ terms. At the maximum degrees this gives 8,007 terms for var1 and 1,770 for var2, both exceeding the number of training rows. Regularization was therefore included to control the coefficients, particularly at higher degrees.

3. Model selection and validation

Three penalty types were compared. Ridge shrinks all coefficients through an L2 penalty; LASSO can set coefficients exactly to zero through an L1 penalty; Elastic Net combines both penalties. Since polynomial terms can be strongly correlated, all three methods were compared on the same validation folds.

Setting	Values used
Polynomial degree	Every integer from 1-10 for var1 and 1-20 for var2, following the assignment limits.
Regularization strength α	Initial grid: 10^k , $k = -6, -5, \dots, 6$. Log spacing covers weak to strong penalties without assuming a useful scale.
Elastic Net L1 ratio ρ	0.1, 0.3, 0.5, 0.7, 0.9 and 0.95. LASSO and Ridge separately cover the pure L1 and L2 alternatives.
Local refinement	Nine log-spaced α values around the best coarse configuration in each family. Boundary winners extend two decades outward; Elastic Net rechecks all six ratios at its selected coarse degree.

For n training rows, the implemented objectives were:

Ridge: $\text{SSE} + \alpha \|\beta\|_2^2$

LASSO: $\text{SSE}/(2n) + \alpha \|\beta\|_1$

Elastic Net: $\text{SSE}/(2n) + \alpha\rho \|\beta\|_1 + \alpha(1 - \rho)\|\beta\|_2^2/2$

Here SSE is the sum of squared training residuals. Because Ridge uses a different loss normalization, numerical α values are tuned separately and are not directly comparable across families. For LASSO and Elastic Net, y was centered using the training-fold mean and that mean was added back to predictions; the target was not variance-scaled.

Nested cross-validation: five outer folds, three inner folds

Inner selection. Within each outer-training partition, the complete degree, model, α and ratio search used three validation folds. MSE was averaged across folds for each configuration separately, not across model families. The lowest eligible mean MSE selected the configuration.

Outer evaluation. The selected pipeline was refitted on the outer-training rows and evaluated once on the held-out outer fold. Repeating this five times separates hyperparameter selection from assessment and reduces the optimism of reporting only the minimum tuning error.

var1 used shuffled KFold. var2 used shuffled GroupKFold with identical coordinate vectors kept together in both loops, so repeated locations could not appear in training and validation simultaneously. Seeds were 175 for inner splits and 176 for outer splits. All final test rows were excluded from fitting and tuning.

$\text{MSE} = \text{SSE}/n$ was minimized; $R^2 = 1 - \text{SSE}/\text{SST}$ was also reported, with SST measuring variation about the validation-fold mean. Final settings were chosen by a separate three-fold search on all labelled rows, followed by a full-data fit. This search ran before outer evaluation; no outer score changed the final choice. The outer results therefore assess the selection procedure, not a single fixed model trained on all 1,000 rows.

4. var1: steam turbine optimization

Final configuration: LASSO, degree 5, $\alpha = 0.01$.

Selection CV: MSE = 0.35519; $R^2 = 0.96840$.

Fitted expansion: 461 polynomial columns, of which 119 have nonzero coefficients.

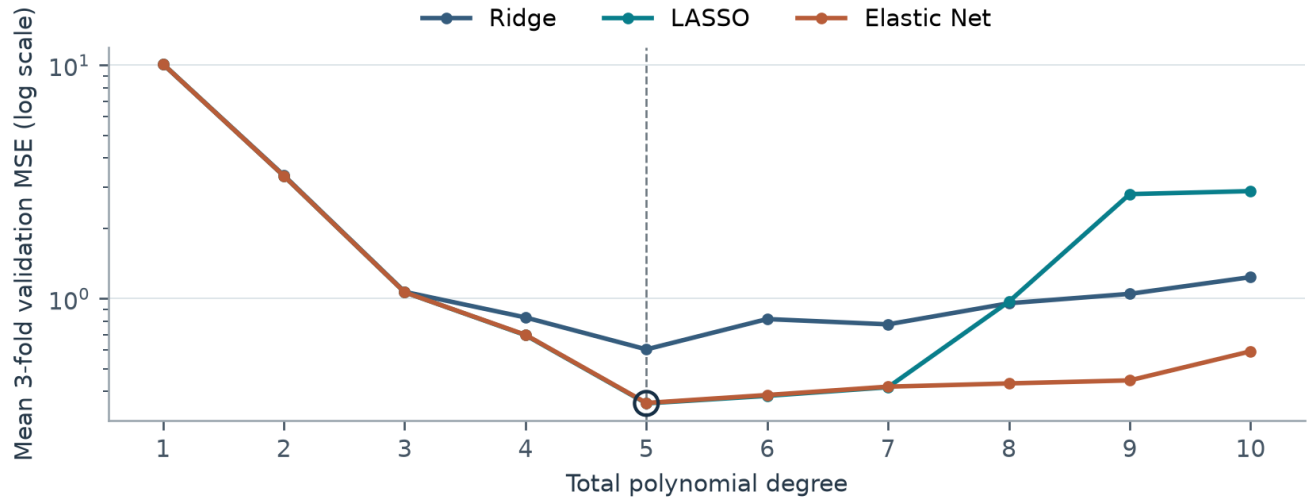


Figure 1. Lowest fully evaluated, converged three-fold MSE at each degree and model family in the final search, including refinement where performed. The vertical axis is logarithmic; the ring marks the selected model. These are tuning scores, not independent outer-fold scores.

Best within each family	Degree	α	L1 ratio	CV MSE
LASSO	5	0.01	1.00	0.35519
Elastic Net	5	0.01	0.95	0.35660
Ridge	5	17.78279	-	0.60518

Discussion

Validation error fell substantially as the degree increased from 1 to 5, while degrees above 5 did not improve the selected score. There was no validation-based reason to use a higher degree. LASSO had the lowest eligible MSE. Of the 461 expanded columns, only 119 retained nonzero coefficients in the final fit.

The near-LASSO Elastic Net configuration was very close: its MSE exceeded LASSO by only about 0.00142. The two scores are too close to claim a clear practical advantage for LASSO. It was selected because its mean MSE was smaller, whereas the best Ridge score was noticeably higher at 0.60518.

Selection was stable across the independent outer runs: all five chose degree-5 LASSO with $\alpha = 0.01$. The mean outer MSE was **0.33740 ± 0.04907** and mean outer R^2 was **0.96985 ± 0.00314**, where $\pm$ is the sample standard deviation across the five folds. The agreement between the outer-fold choices gives more confidence in this selection than the small LASSO-Elastic Net score difference alone.

5. var2: thermal reservoir mapping

Final configuration: Elastic Net, degree 11, $\alpha = 0.001$, L1 ratio = 0.3.

Selection CV: MSE = 0.26367; $R^2 = 0.99397$.

Fitted expansion: 363 polynomial columns, of which 233 have nonzero coefficients.

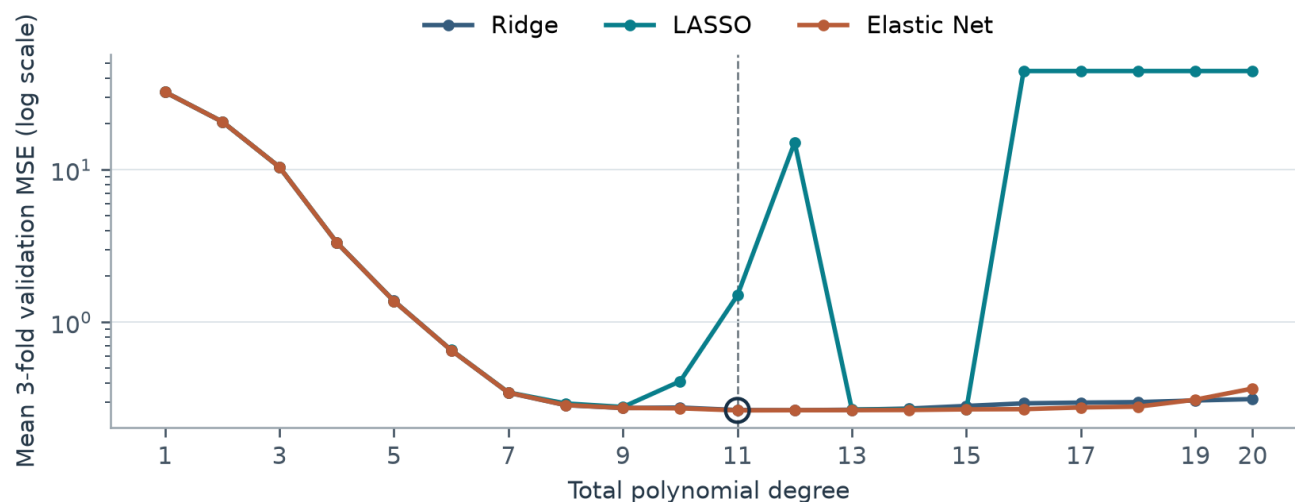


Figure 2. Final-search comparison using the definition in Figure 1. Pruned or numerically unfinished candidates are omitted; LASSO spikes should be read alongside the convergence limits in Section 6. The search extends to degree 20; stopping at 10 would have missed the selected configuration.

Best within each family	Degree	α	L1 ratio	CV MSE
Elastic Net	11	0.001	0.30	0.26367
Ridge	12	1.0	-	0.26427
LASSO	13	0.001	1.00	0.26693

Discussion

The lower-degree fits had larger validation errors. Scores improved at higher degrees, but increasing the degree all the way to 20 did not produce the best result. Degree-11 Elastic Net achieved the lowest eligible mean MSE. Its L1 component permits coefficient selection and its L2 component distributes shrinkage across correlated terms. The chosen ratio of 0.3 places more weight on the L2 component than on the L1 component in the specified objective.

Ridge at degree 12 was almost tied with the winner: the MSE difference was only about 0.00060. The outer searches selected one LASSO, two Ridge and two Elastic Net models, with degrees ranging from 9 to 13. Unlike var1, the selected family and degree changed between outer folds. Degree 11 was retained for the final model because it won the search on all labelled rows; the close Ridge result should also be kept in mind when interpreting this choice.

The mean outer MSE was **0.27880 ± 0.03654** and mean outer R^2 was **0.99344 ± 0.00175**. Grouped validation keeps repeated coordinates out of training when their location is held out. It does not test spatial extrapolation to an entirely new geological region.

6. Validation summary and submitted files

Dataset	Mean outer MSE $\pm$ SD	Mean outer $R^2 \pm$ SD
var1	0.33740 ± 0.04907	0.96985 ± 0.00314
var2	0.27880 ± 0.03654	0.99344 ± 0.00175

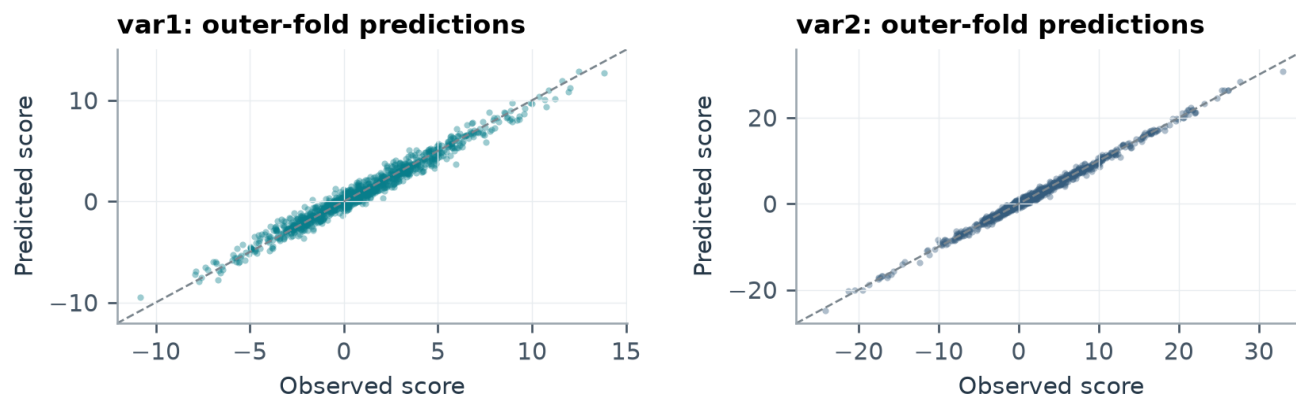


Figure 3. Each point is a training observation predicted when it was held out in an outer fold (1,000 points per dataset). The dashed line denotes perfect agreement. The table reports unweighted fold means; fold SD is descriptive variation, not a confidence interval.

Computation and limitations

Ridge used a shared eigendecomposition across α values, checked against scikit-learn Ridge. LASSO and Elastic Net used accelerated skglm solvers. Coarse fits were limited to 10 working-set iterations and 500 coordinate epochs per working set; promising unfinished candidates and local refinements received longer budgets. Eligibility required a first-order optimality (KKT) residual at most 5×10^{-5} . Final models used tighter fitting settings and passed this check.

The final-search records contained 36 fully evaluated nonconverged rows for var1 and 311 for var2; these could not win. For var2, another 646 rows were stopped when a partial-error lower bound could not beat a fully evaluated leader. Counts include refinement and retry records, not necessarily unique configurations. Some early var1 checkpoints used a larger solver budget but passed the same eligibility threshold. Consequently, “best” refers to the eligible configurations evaluated with the stated grids, tolerances and refinement rule, not every possible polynomial model.

Prediction files and code

Each chosen pipeline was fitted on all 1,000 labelled rows. Predictions are supplied as **BT2024175_pred_var1.csv** and **BT2024175_pred_var2.csv**, each with 1,000 rows, one column named y and no index. Original test order and repeated rows were preserved. Reloaded models reproduced both CSVs; all ten outer-fold scores were independently recomputed from saved predictions. No test targets were supplied, so test-set accuracy is unknown.

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The GitHub repository contains the training and prediction code, saved models, and cross-validation results. Instructions for reproducing the experiments and generating predictions are provided in the README.